

Recall

Exhaustive Capability Packet for Independent AI Review

This document is designed to be uploaded to another AI system. It separates implemented behavior from configuration-dependent behavior, partial UI, known limitations, and roadmap-like concepts.

Product: Recall personal operating system

Production: <https://recall-app.net>

Repository audited: Aura AI Personal Assistant monorepo

Snapshot date: July 18, 2026

Audience: product, architecture, UX, security, and capability reviewers

Instructions for the reviewing AI

Treat this as a current-state capability audit, not a roadmap or marketing document. Preserve the status labels. Do not assume configuration-dependent integrations are enabled in every deployment. Do not infer missing capabilities from the product vision.

Status legend

- **IMPLEMENTED** Working code and a usable product path exist.
- **CONFIG-DEPENDENT** Implemented, but requires credentials, external access, or synced data.
- **PARTIAL** Some behavior exists, but UI, coverage, persistence, or workflow completeness is limited.
- **NOT IMPLEMENTED** Explicitly absent or only represented by a static/decorative UI.

Executive summary

Recall is a single-user-oriented AI personal operating system that captures and syncs information from notes, tasks, files, mail, calendars, contacts, finance, ticket email, and smart-home sources. It converts raw inputs into structured records, connects related entities, provides evidence-backed natural-language answers, and ranks actionable items on Today.

Capture / Connect → Preserve source → Normalize → Extract structure → Human review → Link entities → Retrieve / Ask → Show evidence → Act

The strongest implemented areas are evidence-backed Ask, capture and AI Inbox, notes and attachment extraction, Life Memory, people/waiting-on, project source links, finance summaries, deadline attention, and modular connectors. The largest current product gaps are the static Canvas page, incomplete task/project controls in the visible UI, filename-only quick-capture attachments, text-centric document storage, connector removal UX, and deployment-dependent integrations.

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1. Product model and architecture

IMPLEMENTED Recall uses a React/Vite/TypeScript frontend, Express TypeScript API, PostgreSQL with Drizzle, OpenAI integrations, durable database jobs, and modular connectors. Production is hosted behind nginx with PM2 on DigitalOcean.

Core architectural capabilities

- Shared data model across captures, notes, tasks, people, projects, memories, knowledge, documents, assets, connectors, and evidence.
- Raw external source records are preserved and deduplicated using connector/external identity.
- Entity links create typed relationships such as project-to-mail, project-to-transaction, attention-to-source, attention-to-person, person-to-organization, and receipt-to-transaction.
- Evidence records connect claims and derived items back to source records, excerpts, URLs, and metadata.
- Durable jobs support capture extraction, attachment extraction, Evernote import, digest regeneration, and deadline scanning.
- Hybrid retrieval combines full-text/keyword search, semantic embeddings, structured intent handlers, and live connector search.
- External systems generally remain source of truth; Recall stores searchable normalized snapshots and provenance.

2. Home and Ask

IMPLEMENTED Conversational Home
• Central natural-language question input.
• Microphone dictation where browser speech recognition is available.
• Answer read-aloud using supported speech synthesis.
• Past-question history and handoff to the full Ask view.
• Contextual suggestion prompts derived from current Recall data.
• Related source images when answers are grounded in image evidence.
IMPLEMENTED Evidence-backed Ask
• Streaming AI answers with deterministic fallback behavior when AI is unavailable.
• Threaded conversations and follow-up context.
• Search across notes, attachments, Life Memory, Knowledge, people, tasks, projects, documents, captures, and synced source records.
• Specialized intent handling for Gmail, Drive, finance, notes, waiting-on, people, and Homey.

- Confidence, caveats, related records, source metadata, evidence excerpts, and privacy details.

- Durable Ask threads preserve the evidence snapshot used for each answer.

- Thumbs-up/down answer feedback.

- Per-user Ask rules injected as standing instructions.

- Optional speech playback and answer images.

CONFIG-DEPENDENT Live external search

- Live Gmail query across connected Google accounts for mail-oriented questions.

- Live Drive search using Google's full-text index, including content indexed by Google OCR.

- Finance and Homey questions require connected/synced data or a live integration.

- OpenAI synthesis, embeddings, vision OCR, and OpenAI TTS require configured AI credentials.

3. Today and proactive attention

IMPLEMENTED Ranked action queue

- Combines recommended focus, urgent tasks, AI Inbox reminders, waiting-on items, Homey alerts, and attention deadlines.
- Deduplicates overlapping recommendations.
- Links each queue entry to the underlying entity or source context.
- Allows waiting-on dismissal and follow-up task creation.
- Allows deadline Seen, Dismiss, and one-day-before Snooze actions.
- Supports quick brain-dump capture into note, task, or AI Inbox flows.

IMPLEMENTED Briefing and digest

- Daily briefing, focus recommendation, “don’t forget” list, and Homey urgency integration.
- Weekly digest with structured sections and local dismissal state.
- Finance snapshot and waiting-on context can contribute to Home/Today.

4. Capture and AI Inbox

IMPLEMENTED Capture channels

- Manual typed capture and dictated capture.
- Today/Home brain dump.
- Browser extension page/selection capture.
- Connector ingestion, document import, attachment upload, Evernote import, and Life Memory import.
- Offline client queue with retry for manual capture submission.
- Raw capture is retained for provenance.

IMPLEMENTED AI extraction and review

- Versioned classification into note, task, reminder, work note, project item, or reference.

- Suggested cleaned title, priority, absolute due date, project, tags, actions, requester, confidence, and evidence quote.

- Human edit/correction before acceptance.

- Accept into notes, tasks, or Life Memory; dismiss irrelevant suggestions.

- Person correction/resolution and source evidence preservation.

- Durable processing status, retry, and failure visibility.

PARTIAL Quick Capture file selection currently appends filenames to capture/note text; it does not upload those selected files.

5. Notes and Notebooks

IMPLEMENTED

- Create, load, edit, search, pin, and delete notes.
- Create notebooks and assign notes to notebooks.
- Tags, person tagging, project association, and note metadata.
- Keyword/full-text search and semantic search.
- AI note summary and contextual AI writing/chat behavior.
- Microphone dictation where supported.
- Rich imported Evernote content, including embedded images where supported.
- Large/chunked ENEX imports with durable progress and resume behavior.
- Attachments with durable extraction jobs.
- Text extraction from text/PDF-like content and vision OCR for images.
- Ask and Notes search include extracted attachment text after processing.

PARTIAL Some richer note/task operations exist in API/context code but are not exposed uniformly in the main Notes UI.

6. Tasks

IMPLEMENTED

- Quick task creation.
- Complete and reopen tasks.
- Assign a person/requester and filter by person.
- Show task evidence where available.
- Create follow-up tasks from waiting-on recommendations.
- Task-aware AI chat with optional voice.
- API support includes broader update/delete and structured task fields.

PARTIAL Upcoming/All/Filter tabs, drag handle, overflow menu, and paperclip are decorative or incomplete. The visible page does not expose the API's full edit/delete surface.

7. Life Memory

IMPLEMENTED Life Memory stores durable personal facts that should rank above transient notes.

- Create categorized facts across family, home, vehicles, health, work, finance, preferences, and other domains.
- AI-assisted classification and conservative extraction.
- Search, filter, pin, edit, archive, and inspect history.
- Corrections preserve history using version supersession rather than destructive overwrite.
- Effective and expiry dates for time-bound facts.
- Evidence/source metadata attached to memories.

- Import review flow before records become active.
- Markdown “Life File” export.
- Active/pinned memories are prioritized in Ask retrieval.

8. Knowledge

IMPLEMENTED

- Create, edit, delete, search, and categorize reusable knowledge items.
- Supported concepts include procedures, references, contacts, snippets, instructions, and checklists.
- Person tagging and evidence metadata.
- Knowledge items participate in Ask retrieval.

9. People and waiting-on

IMPLEMENTED

- Contact create/read/update/delete and search.
- Profile details and person-centric navigation.
- Primary-person links from notes, tasks, captures, and attention items.
- Person timelines and related notes/tasks/knowledge/memories.
- Duplicate-person merge with foreign-key/link reassignment and alias preservation.
- Automatic waiting-on heuristics across tasks, notes, captures, and Gmail.
- Durable waiting-on dismissals.
- Create a follow-up task directly from a waiting-on item.

10. Organizations, vendors, and invoices

IMPLEMENTED

- Create, edit, delete, search, and view organizations.
- Organization classifications include vendor/contractor-oriented use.
- Typed person-to-organization affiliation links.
- Organization-associated records and activity context.
- Invoice create/update/status tracking tied to organizations.
- Invoice amounts, dates, descriptions, and related records.

11. Projects and timelines

IMPLEMENTED

- Create projects and open a project detail page.
- Group related notes, tasks, and captures.
- Project context cards and subject timelines.
- Search synced Gmail and finance records from project detail.
- Confirm link/unlink of mail and transactions using typed entity links.
- Navigate from project context to original source records.
- Project-name keyword hints support conservative linking suggestions.
- API supports broader project patch/status behavior.

PARTIAL The visible project UI emphasizes creation, detail, timeline, and source linking. Project edit/archive controls are not fully exposed.

12. Documents and attachments

IMPLEMENTED

- Import text, CSV, Markdown, JSON, and similar content.
- CSV preview and structured metadata.
- AI document summaries.
- Search/cite stored document text in Ask.
- Evidence links and source navigation.
- Receipt-to-transaction matching with confirmation workflow.
- Note attachments support background text extraction and image OCR.

PARTIAL The Documents module primarily stores extracted text and metadata, not original binary content. Unsupported binaries may be stored by filename only. Client-side text imports are bounded to approximately 200 KB.

13. Finance intelligence

CONFIG-DEPENDENT Requires a configured finance API connector and successful sync.

- Normalized transaction ingestion with stable external IDs and source provenance.
- Deterministic spending, income, and net calculations.
- Date-range, category, payee/merchant, account, project, vehicle, and home analysis.
- Transaction breakdowns behind reported totals.
- Classification of expense, income, transfer, credit-card payment, and refund.
- Transfer and credit-card-payment exclusion from spending totals to avoid double counting.
- Payee/category summaries and finance snapshot.
- Likely-subscription detection from recurring transactions.
- Receipt matching and confirmable transaction links.
- Project/asset spending suggestions and confirmed typed links.
- Ask finance intents use records and deterministic aggregation rather than model arithmetic.

NOT IMPLEMENTED Recall is not the authoritative ledger and is not a high-stakes financial advisor. Budget editing and financial write-back are not core current behavior.

14. Vehicles, homes, and warranties

IMPLEMENTED

- Vehicle profiles with identifying/metadata fields such as VIN.
- Home profiles with address and descriptive metadata.
- Warranty CRUD linked to vehicle, home, or other subject.
- Warranty provider, expiry, notes, and status context.
- Subject timelines across linked records.

- Spend rollups and transaction suggestions for assets.
- Confirm or unlink asset-to-transaction relationships.

15. Deadline Attention System

IMPLEMENTED

- Durable attention records with title, summary, due date, kind, status, source, evidence, person/project links, confidence, and metadata.
- Status lifecycle: Open, Seen, Snoozed, Dismissed, Completed.
- Google Calendar future events promoted without an LLM when start metadata exists.
- Gmail dated-commitment extraction using a versioned structured prompt.
- High-confidence auto-create threshold; explicit dated commitments only.
- Source/person entity links and source evidence.
- Deduplication by user, source entity, source ID, and due date.
- Today ranking: overdue, due soon, snooze-expired, then new unseen.
- API actions for Seen, Dismiss, Snooze presets/custom date, and Complete.
- Automatic attention scan job after Google sync and a manual API scan endpoint.

PARTIAL Today exposes Seen, Dismiss, and one-day-before Snooze, not every API action/preset. No visible manual “scan now” control was found.

NOT IMPLEMENTED Outbound email, SMS, and push reminder delivery.

16. Connectors

Connector	Status	Current capabilities	Important limits
Google	Config-dependent	Multi-account OAuth; Gmail, Calendar, Contacts, Drive read-only sync; live Gmail/Drive search; attention scan after sync.	Cached sync is bounded. Drive OCR may remain in Google's index rather than being downloaded as raw text.
Microsoft	Config-dependent	OAuth; recent Outlook mail and Teams chat/message snippets.	Bounded recent data. Tenant policy may block Teams access.
Finance API	Config-dependent	Transaction fetch, normalize, dedupe, classify, summarize, and evidence mapping.	Often environment/bootstrap configured; no ledger write-back.
Homey Pro	Config-dependent	OAuth; device/flow sync; Ask status/control; flow triggering; webhook alerts.	Requires Athom app credentials and supported device capabilities.
Ticket email	Config-dependent	IMAP connection; ticket email parsing into source records and follow-up context.	Requires accessible IMAP credentials/server.

CSV	Implemented	Normalize/import tabular source records with stable connector identity.	Field quality depends on imported data.
Manual capture	Implemented	Freeform capture, source metadata, AI Inbox extraction.	AI extraction may require review.
Evernote ENEX	Implemented	Chunked import of notes and supported attachments with durable jobs.	Format edge cases and unsupported attachments may be partial.
Browser extension	Implemented	Capture selected/current-page text, URL/title, Outlook/Teams context, retry queue.	Queue retries when popup is used again; no continuous background flush.

IMPLEMENTED Sync runs record fetched, created, updated, failed, completion, and error states. Source records are deduplicated. Connector failures are isolated.

PARTIAL Connectors UI emphasizes connect/test/sync; no complete disconnect/delete flow is exposed.

17. Homey smart-home capabilities

CONFIG-DEPENDENT

- OAuth connection and encrypted credentials.
- Device and Flow inventory sync into source records.
- Ask queries for device status.
- Explicit capability control and supported Flow triggering.
- Authenticated webhook endpoint for Homey Flow alerts.
- Alert severity, acknowledgement, Today/Urgency ranking, and quiet-hour handling for informational alerts.
- Webhook secret rotation, webhook detail display, and test alert action.
- Documented recipes for smoke, leak, and prolonged-open-door alerts.

18. Browser extension

IMPLEMENTED Chrome/Edge Manifest V3 extension.

- Capture selected text or current-page visible text.
- Capture URL, title, hostname, timestamp, and contextual metadata.
- Specialized Outlook and Teams page-context collection.
- Preview/edit before sending.
- Capture-only revocable token; cannot access notes, Ask, finance, connectors, or settings.
- Token stored client-side while only its SHA-256 hash is stored by Recall.
- Offline/failure queue with retry during later popup use.

NOT IMPLEMENTED A continuously running service worker that flushes queued captures without opening the popup.

19. Activity, privacy, security, and administration

IMPLEMENTED Activity and audit

- Activity history for captures, acceptance, sync, Ask, and other actions.
- Entity/source navigation from activity where applicable.
- Durable Ask evidence and privacy snapshots.

IMPLEMENTED Authentication and account security

- Login and deployment-policy-controlled registration/invitation behavior.
- Bcrypt password hashing.

- Revocable HttpOnly server sessions.
- Password change with session revocation behavior.
- CORS restrictions and endpoint rate limiting.
- Encrypted connector credentials server-side.
- Narrow, revocable, expiring browser-extension tokens.
- Privacy Policy and Terms pages.

IMPLEMENTED Administration

- List users.
- Enable/disable user access.
- Grant/remove administrator role.
- Reset a user password.
- Manage personal Ask rules separately from administrator controls.

20. Data, evidence, and AI behavior

Evidence model

- Evidence can identify entity, claim type, source record, excerpt, URL, row/page metadata, and additional metadata.
- Connector sync updates current evidence rather than appending duplicates indefinitely.
- Legacy duplicate evidence is collapsed for display.
- Ask stores exact excerpts and source metadata used at answer time.
- Financial totals expose their period and underlying transaction rows.

AI behavior

- Versioned prompts for important structured extraction.
- Structured JSON validation and deterministic fallbacks.
- Confidence/caveat handling.
- Conservative person, project, due-date, and deadline extraction.
- AI is used for synthesis, semantic matching, classification, summarization, and OCR—not deterministic finance arithmetic.
- User corrections are preserved and may guide later behavior.
- Ask rules provide user-defined standing instructions.

Core entity families

Users, sessions, notebooks, notes, note attachments, captures, capture items, tasks, people, projects, organizations, invoices, vehicles, homes, warranties, Life Memory, Knowledge, documents, connectors, sync runs, source records, evidence, entity links, entity embeddings, Ask threads/messages, jobs, waiting dismissals, user rules, corrections, attention items, and audit/activity records.

21. Consolidated partial features and gaps

Area	Status	Current limitation
Canvas	Not implemented beyond mockup	Static visual page; nodes, editing, zoom, minimap, AI input, share, and persistence are not wired.
Tasks UI	Partial	Several controls are decorative; full edit/delete and richer fields are not consistently exposed.
Projects UI	Partial	Creation/detail/source linking work; edit/archive controls are not fully surfaced.
Quick Capture files	Partial	Selected filenames are appended; files are not uploaded through this flow.

Documents	Partial	Text/metadata-centric; original binaries are not generally retained by the Documents module.
Connector removal	Partial	No complete disconnect/delete management UI.
Finance creation	Partial/config	Finance connector is typically environment/bootstrap driven rather than created from visible UI.
Microsoft	Bounded/config	Recent bounded mail/chat data; tenant policy can block Teams.
Google cache	Bounded/config	Local sync is bounded, offset by live Gmail/Drive query paths.
Attention UI	Partial	Not all server actions/presets or manual scan are exposed visibly.
Outbound reminders	Not implemented	No SMS, push, or reminder email delivery.
Extension retry	Partial	Retry depends on reopening/using the popup.
External write-back	Mostly not implemented by design	Most connectors are read-only; Homey explicit control is the principal exception.

22. Representative end-to-end workflows

A. Capture a work request

Dictate/paste → Raw capture → AI classification → Review person/date/priority → Accept task → Today/People surface follow-up → Evidence returns to source

B. Lawyer email with court date

Google sync → Gmail source record → Structured deadline extraction → Confidence gate → Attention item + person/source links → Today → Seen/Snooze/Dismiss → Ask cites email

C. Find a fact in a scanned document

Attach image/PDF → Durable extraction/OCR → Indexed text → Ask/Notes search → Answer + evidence excerpt/source

D. Analyze project spending

Finance sync → Normalize/classify → Suggest/confirm project links → Deterministic aggregation → Ask answer → Inspect transactions

E. Track something owed by another person

Task/note/capture/Gmail → Waiting-on heuristic → People/Today → Dismiss or create follow-up task

F. Home emergency alert

Homey Flow → Authenticated Recall webhook → Alert severity → Today/Urgency queue → Acknowledge → Source/device context

G. Teach Recall a permanent fact

Create/import capture → Life Memory review → Categorize/pin/evidence → Ask priority → Correct via versioned supersession → Export Life File

23. Code evidence map

Paths are repository-relative.

Capability area	Primary implementation evidence
Application routes/pages	artifacts/recall-app/src/App.tsx; artifacts/recall-app/src/pages/
Home and Ask UI	pages/Dashboard.tsx; pages/Ask.tsx

Ask retrieval/reasoning	<code>api-server/src/services/query-engine.ts</code> ; retrieval, Gmail, Drive, finance, Homey services
Today	<code>pages/Today.tsx</code> ; <code>components/home/TodayActionQueue.tsx</code> ; <code>routes/today.ts</code>
Capture/Inbox	<code>components/CaptureModal.tsx</code> ; <code>pages/Inbox.tsx</code> ; <code>routes/captures.ts</code> ; capture pipeline services
Notes/attachments	<code>pages/Notes.tsx</code> ; <code>routes/notes.ts</code> ; <code>routes/attachments.ts</code> ; attachment extraction service
Tasks	<code>pages/Tasks.tsx</code> ; <code>routes/tasks.ts</code>
Memory/Knowledge	<code>pages/Memory.tsx</code> ; <code>pages/Knowledge.tsx</code> ; <code>routes/memory.ts</code> ; life-memory service
People/Organizations	<code>pages/People.tsx</code> ; <code>pages/Organizations.tsx</code> ; corresponding API routes/services
Projects	<code>pages/Projects.tsx</code> ; <code>pages/ProjectDetail.tsx</code> ; <code>routes/projects.ts</code> ; project source/timeline services
Documents	<code>pages/Documents.tsx</code> ; <code>routes/documents.ts</code> ; receipt match services
Finance	<code>services/finance-sync.ts</code> ; <code>connectors/finance-api.ts</code> ; finance classification/aggregation services
Vehicles/homes/warranties	<code>pages/Vehicles.tsx</code> ; <code>routes/vehicles.ts</code> ; subject spend/timeline services
Deadline attention	<code>routes/attention.ts</code> ; <code>services/attention*.ts</code> ; <code>prompts/extractDeadline.v1.ts</code>
Connectors	<code>pages/Connectors.tsx</code> ; <code>routes/connectors.ts</code> ; <code>api-server/src/connectors/</code>
Homey	<code>connectors/homey.ts</code> ; Homey services; <code>routes/webhooks-homey.ts</code> ; <code>docs/Homey_Flow_Cookbook.md</code>
Browser extension	<code>artifacts/recall-extension/manifest.json</code> ; <code>popup.js</code>
Security/auth	<code>routes/auth.ts</code> ; <code>middleware/security.ts</code> ; <code>lib/secret-box.ts</code> ; session/extension-token services
Database schema	<code>lib/db/src/schema/</code> ; <code>lib/db/migrations/</code>

24. Suggested questions for the reviewing AI

1. Which implemented workflows create the most user value, and which are difficult to discover?
2. Where do the current UI limitations prevent the backend capabilities from being useful?
3. Which capabilities overlap or create competing mental models?
4. Does Today rank the right categories, and how should Seen/Snooze/Complete differ?
5. Is the evidence model sufficient for high-trust personal, legal, and financial questions?
6. Which connector behaviors should remain read-only, and where is write-back justified?
7. What privacy, security, prompt-injection, or data-retention risks deserve priority?
8. How should Recall communicate sync freshness, extraction confidence, and missing data?
9. Which partial features should be completed, removed, or hidden?
10. What is the clearest product information architecture for Home, Today, Ask, Memory, Notes, and Knowledge?
11. What test/evaluation suite should measure answer grounding, deadline precision, finance correctness, and waiting-on quality?
12. What three milestones would most improve reliability and daily usefulness?

Final scope statement

This packet aims to enumerate all material implemented capabilities and all material known limitations in the audited repository as of July 18, 2026. It is more exhaustive and technically explicit than the separate user-facing app guide. It intentionally does not list every database column, every API parameter, or every decorative UI element.

Prepared from repository documentation plus direct audit of routed frontend pages, API route families, services, connectors, jobs, security implementation, and browser-extension code. No secrets or credentials are included.